

Comparison of Hydrodynamic Responses of Different Cage Structures in the Integrated Photovoltaic and Fish Aquaculture System

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Abstract: As an important means of conserving fishery resources and restoring ecological environments, marine aquaculture has developed rapidly in recent years. However, research on the structural analysis of aquaculture cages remains scarce. In this study, we present a new design of an integrated photovoltaic and fish aquaculture system. Specifically, this study investigates the hydrodynamic responses of rectangular cages, octagonal cages, and hexadecagonal cages for the system based on the Navier-Stokes equations and the continuity equation, including the external forces on the cage facades and internal flow velocities. The results shows that the flow velocity distribution inside the surface hexadecagonal prism is better. The article also concludes that increasing the number of sides in the cage design can better support fish welfare, which is another novel discovery.

Keywords: aquaculture; fish cage; structure; internal flow velocity; Navier-Stokes;

1. Introduction

Recently, aquaculture has become the main source of seafood production. According to the Food and Agriculture Organization of the United Nations, global aquaculture output has surged dramatically over the past two decades: from 43 million tons in 2000, it reached 130 million tons by 2022, marking a 204% increase and an average annual growth rate of 5.2% [1]. This expansion is encouraged by the advantages of offshore aquaculture, which can provide higher water quality and a wider activity space for fish [2]. Additionally, The aquaculture facilitates better regulatory oversight and quicker responses to environmental changes or disease outbreaks, enhancing sustainability [3].

1.1 Structural Foundations for Aquaculture

The fundamental prerequisite for realizing deep-sea and offshore aquaculture lies in safe and reliable equipment, including but not limited to net cages, enclosures, platforms, and factory ships. Due to the characteristics of the marine environment, deep-sea and offshore aquaculture equipment are generally required to possess properties such as wind and wave resistance, disaster avoidance, and self-preservation capabilities. Therefore, many types of aquaculture platforms have been designed, representative structures include floating deep-water net cages, column-type net cages, and semi-submersible net cages. Despite the diversity of structures, classification societies and similar institutions in various countries have not yet issued mature standards or guidance documents for deep-sea aquaculture net cages. Most of them are based on the standards for offshore mobile platforms or fixed facilities, and there is relatively little research on deep-sea aquaculture net cages.

Meanwhile, the hydrodynamic characteristics inside fish cages largely affect fish behavior and aquaculture profits. For example, in strong currents, salmon consume more energy for swimming, which directly affects aquaculture welfare [4]. In certain wave structures or fluid movements, fish may also be unable to avoid collisions with each other, which increase losses[5]. Therefore,

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studying the hydrodynamic response of net cages is of great significance to the aquaculture industry.

1.2 Sustainable Energy Supply

With the development of aquaculture technology, the aquaculture industry is becoming increasingly intelligent. Leveraging advanced smart equipment, the aquaculture accurately monitors key parameters such as water temperature, water quality, and tides. This not only enhances breeding efficiency and significantly improves the quality of aquaculture products but also greatly increases fishery yields. Inevitably, these improvements in efficiency and yields has led to a greater demand for energy. While abundant fish farms offer ideal locations for renewable energy systems. The combination of sustainable energy-generating facilities and offshore aquaculture structures can not only provide a reliable and self-sufficient power supply to fulfill the needs of on-board sensors and other equipment but also offer a promising solution to climate change issues and the urgent need for sustainable energy sources.

Among all sustainable energy forms, wind and solar sources are the most promising approach for offshore locations. To date, the installation of traditional wind turbines and mature and well-established. However, the construction, transport, and site installation of foundations account for approximately 50% of the total development cost of an offshore wind farm [6]. As the number of the wind turbines increases, this proportion will increase simultaneously and significantly [7].

On the other hand, solar energy is also extremely viable and abundant, making it an excellent choice for deployment on aquaculture facilities. Currently, there are not many trials or designs implemented to allow offshore structures to accommodate photovoltaic equipment in harsh and vigorous offshore environments. There are now concept designs for offshore photovoltaic bases, such as those by RWE and other companies [8]. In 2024, China's CHN Energy launched the first and largest offshore photovoltaic facility in Shandong Province, covering an area of approximately 1,223 hectares with a potential total capacity of one gigawatt [9]. For now, the development and application of offshore photovoltaic technology are constrained by deployment costs, which is why the industry remains largely in the stage of concept design and preliminary trials. Integrating fish farms with offshore solar farms provides a low-cost yet effective way to accommodate such technology in the ocean. This concept would bring unprecedented vitality to the development of both industries.

2. Goals

In this study, we compare the different hydrodynamic responses of three cage structure for an integrated photovoltaic and fish aquaculture system. We have chosen the Semi-Submersible rigid cage as a base for the photovoltaic system. Our research investigates and compares the hydrodynamic responses of three possible fish cage shapes: rectangular, octagonal, and hexadecagonal, in integrated photovoltaic and offshore aquaculture systems. It aims to facilitate a deeper understanding of aquaculture cages, and, considering the industry's growing energy demand and the potential of integrating renewable energy, provide sufficient guidance and support for the deployment and commercialization of these integrated systems.

The research consists of three major phases. First, we design the system and model the cages using Creo. Then, we import the model into Comsol to employ CFD based on the Navier–Stokes equations and the continuity equation to explore the pressure on fish cage bodies, analyze the velocity profiles to study fluid behavior, and reveal the impacts of flow fields on fish welfare. Finally, we analyze the hydrodynamic responses and draw conclusions.

3. Materials and Methods

3.1 Material Background

3.1.1 Creo

We use Creo to design a 3D model of the fish cage. Creo is a computer-aided design software developed by PTC [10]. It supports complex assembly design and can accurately define component relationships and constraints. Combining advanced functionality with emerging technologies, Creo offers interoperability, an open architecture, and user-friendly operation, making it suitable for the entire product development cycle—from conceptual design to manufacturing. It is applied in automotive components, aerospace systems, medical devices, and so on [11].

Pramod et al. used Creo for terrain modeling and developed a CFD model to predict and analyze wind resources [12]. Dheeman et al. completed the 3D modeling of a piston valve using Creo's reverse modeling function and analyzed the fluid movement in the working chamber of the damper [13]. Joshi et al. created a 3D model of a propeller with Creo and conducted an analysis of structural mechanical characteristics based on finite element analysis [14].

3.1.2 *Fluent*

After modeling the fish cages in Creo, we then import the files into ANSYS Fluent or Comsol Multiphysics, which are computational fluid dynamics (CFD) software where we conduct the simulation of the system in ocean conditions. ANSYS Fluent is one of the most advanced and widely used CFD software tools in the world. It was founded in 1980 by Fluent Inc., a company founded in Lebanon, New Hampshire, USA, by engineers and scientists from the Hanover area, including former Dartmouth College researchers. Fluent Inc. focused on creating CFD tools based on the finite volume method to meet growing industrial demand for flow simulation in aerospace, automotive, and energy sectors [15]. In 2006, Fluent Inc. was acquired by ANSYS Inc., a global leader in engineering simulation, which integrated Fluent into its ANSYS Workbench platform, thereby enhancing its usability and interoperability with other simulation tools [16].

Since then, ANSYS Fluent has continued to evolve, offering support for increasingly complex physics such as turbulence modeling, multiphase flow, reacting species, and heat transfer. It provides both a graphical user interface (GUI) and scripting capabilities via journal files and Python-based automation (PyFluent). Users can customize simulations using user-defined functions (UDFs), and the software supports both structured and unstructured meshes with dynamic mesh adaptation [17]. Fluent is also optimized for high-performance computing, enabling large-scale simulations across multiple processors or cloud platforms [18]. Today, ANSYS Fluent remains a cornerstone CFD tool used in academic research, product development, and safety analysis across a variety of engineering fields.

ANSYS Fluent has also been successfully used in many peer-reviewed studies. For example, Zhang et al. used FLUENT to implement the forced motion of the aquaculture fishing vessel, and employed CFD based on Navier–Stokes equations to explore how fluid viscosity and nonlinear effects influence the internal and external flow fields of culture compartments, revealing the impacts of flow fields on fish welfare [19]; Zhu et al. used Fluent to study the hydrodynamic performance of vertical-axis turbines [20]; Wang used Fluent to calculate drag forces and coefficients for normalized male and female body shapes at different velocities, and analyzing the effects of varying hip and chest dimensions on drag [21–22]; Moustafa Elkolali et al. used Fluent to optimize the hydrodynamic forces and angle of attack of a small underwater glider, determining the best angles based on the highest lift-to-drag ratio at different speeds [23].

Additionally, other peer-reviewed research has proven that ANSYS Fluent can be used in conjunction with Creo; for instance, Himanshu Sharma et al. used ANSYS Fluent (within ANSYS 19) to perform CFD simulations on triangular prismatic bluff bodies with and without a splitter plate, analyzing drag coefficients and velocity profiles to study fluid behavior around them [24].

3.1.3 *Comsol*

Comsol Multiphysics is also a powerful mathematical modeling software for multiple mathematical and engineering purposes. It supports a broad range of engineering and scientific applications, including heat transfer, fluid flow, electromagnetics, chemical reactions, and structural me-

chanics [25]. It specializes in Multiphysics coupling, allowing users to simulate circumstances involving complex physical fields and offering the opportunity to work with interacting physical effects. For example, in marine and offshore engineering, Comsol can model heat transfer in subsea pipelines, structural deformation under hydrodynamic loading, and corrosion in seawater environments—all within a unified framework [26].

Comsol Multiphysics uses the finite element method for numerical analysis and can also be used in conjunction with coding software such as MATLAB or Java. It is commonly applied in academic research and product development across disciplines including microfluidics, MEMS, thermoelectrics, and offshore energy systems, making it an excellent tool for our research topic.

Comsol Multiphysics has also been successfully used in other peer-reviewed studies. For example, Mina A. Nagi et al used Comsol Multiphysics to perform numerical simulations on two microchannel designs for hydrodynamic focusing, comparing the effects of vertical side channel locations on focusing width, focusing identity, and ion concentration distribution, and investigating how different parameters influence focusing width and concentration profiles [27]; Liu et al employed Comsol Multiphysics to model and analyze the energy capture characteristics of a cylindrical triboelectric nanogenerator (C-TENG), demonstrating its capability to achieve a power density of 30 mW/m²-sufficient to meet the 10 mW power requirement of ocean buoys [28]; Cristian Inostroza's Comsol Multiphysics-based monophasic analysis of raceway bioreactor bends showed that the inclusion of deflectors in baffle partition bends optimized fluid velocity [29].

There are also preexisting peer-reviewed research that combines Creo and Comsol. Kabirnajafi et al. modeled an additively manufactured geothermal heat exchanger and used Comsol Multiphysics to conduct numerical simulations, analyzing three key thermal-fluidic characteristics: wall temperature distribution, bulk fluid temperature variation, and velocity profiles along the helically coiled flow channels. These simulations provided critical performance validation data for the heat transfer capabilities of the innovative screw-type design [30]. These successful use of Comsol multiphysics and the successful combination of Creo and Comsol multiphysics can therefore justify the capability of conducting research like this one using this method and these materials.

In this study, we have chosen Comsol Multiphysics to conduct the research because we have access to a full trial version provided by Comsol China, while we only have access to the student version of ANSYS Fluent, which has a limited number of nodes available. Additionally, Comsol Multiphysics is more flexible and has a user interface that is easier to navigate. Moreover, unlike ANSYS Fluent, which is primarily a CFD software, Comsol Multiphysics not only provides CFD functions but also multiphysics features, enabling better simulations of solid-fluid interactions.

3.1.4 ChatGPT

During the process, ChatGPT, the large language model (LLM) developed by OpenAI, was used for data calculation revision and phrasing revision. ChatGPT's "image generator" (DALL-E) was used to generate two concept design demonstrations for the integrated photovoltaic and fish aquaculture system. As stipulated in the course syllabus, the usage of AI tools like ChatGPT is allowable as long as their application is fully acknowledged, which aligns with ethical research practices.

Furthermore, in accordance with the guidelines established by the Committee on Publication Ethics (COPE) [31]—which emphasize responsible AI usage, human oversight, and full transparency—our applications of ChatGPT and DALL-E strictly adhere to ethical research standards. All AI-assisted content undergoes validation to ensure it does not compromise academic integrity.

ChatGPT can help researchers improve research efficiency to a certain extent. For instance, Zheng et al. used ChatGPT for automating data extraction and enabling downstream predictive modeling [32].

3.2 Tutorial Replicate test

To test the functionality of our tools and prove that they are capable of conducting the study, we have chosen to replicate a tutorial demonstration project provided by Comsol. We start with a

2-dimensional simulation using Comsol Multiphysics to verify its functions. This example demonstrates techniques for modeling fluid–structure interactions in Comsol Multiphysics. The Navier–Stokes equations used to solve the flow are formulated for these moving coordinates [33].

In the original tutorial, Comsol Multiphysics is used to produce graphs that demonstrate fluid flow, graphs that demonstrate mesh velocity (arrows) (see Figure 3.2-1), and mesh and geometry deformation at $t = 0.15$ s (see Figure 3.2-2), and so on.

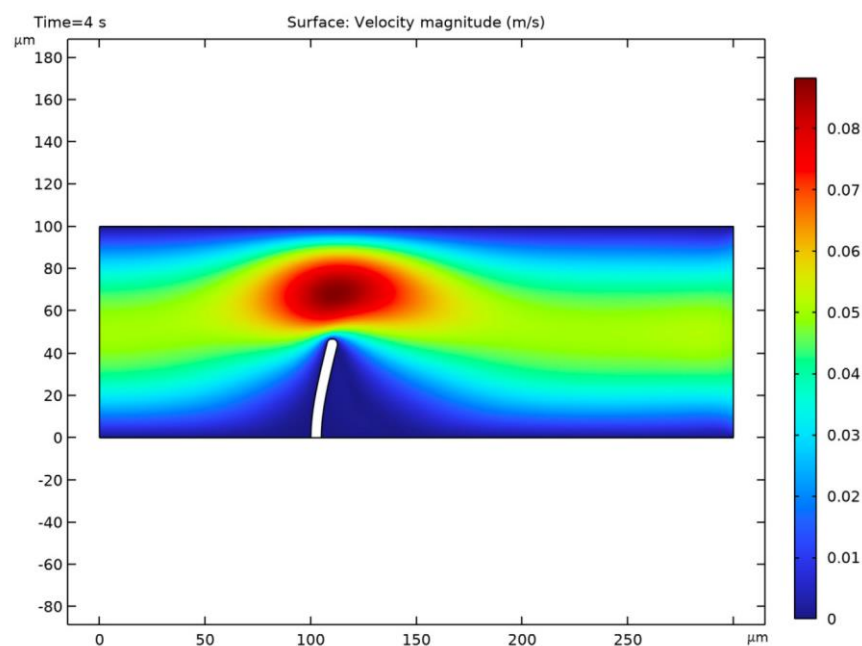


Figure 3.2-1 (from the tutorial (<https://www.comsol.com/model/fluidstructure-interaction-361>))

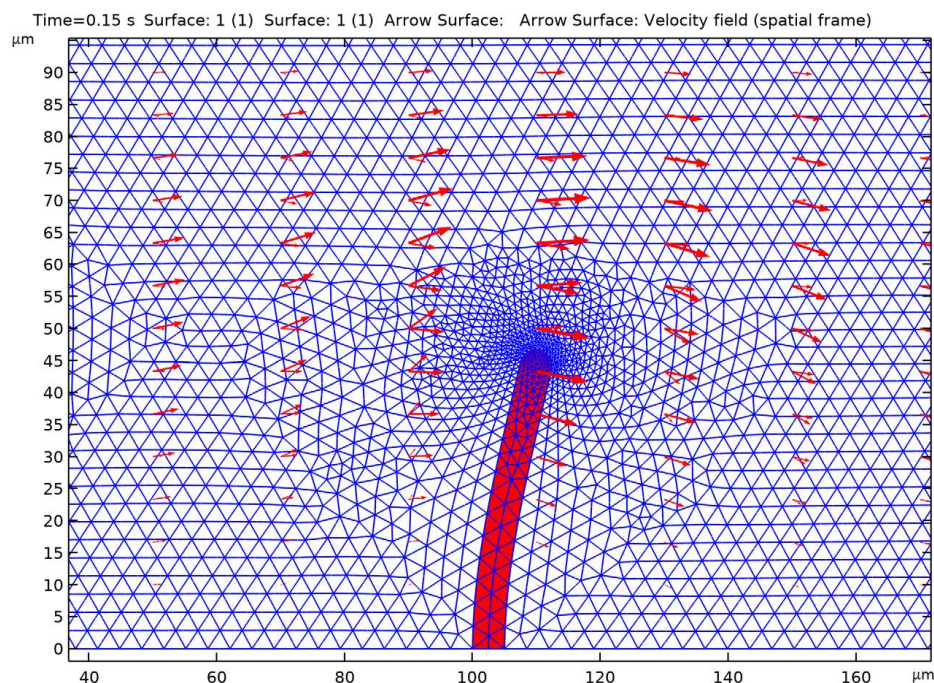


Figure 3.2-2 (from the tutorial (<https://www.comsol.com/model/fluidstructure-interaction-361>))

In our replication test of the tutorial, we successfully produced the distribution of fluid flow and its velocity, and we also generated graphs for these. Additionally, we produced graphs demonstrating the moving grid, mesh velocity (arrows), and mesh and geometry deformation at $t = 0.25$ s

(see Figure 3.2-3), in contrast to $t = 0.15$ s in the tutorial. Furthermore, we have produced a graph for the pressure results (see Figure 3.2-4) .

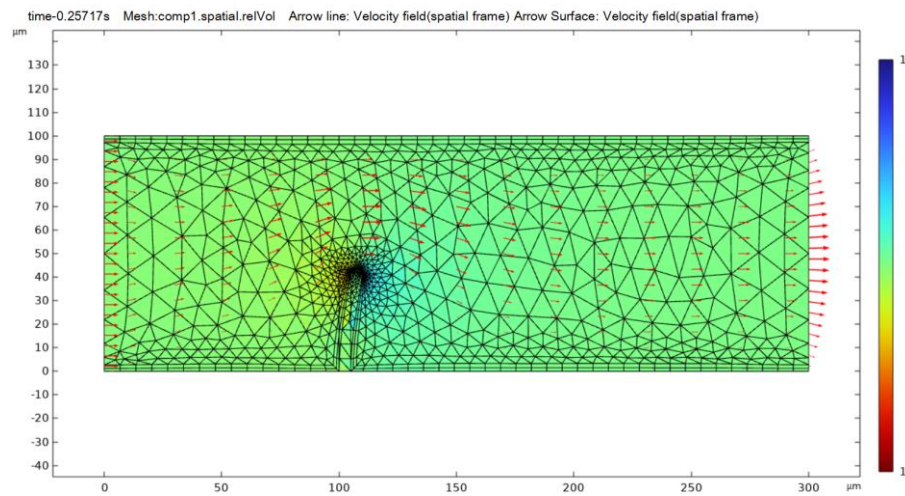


Figure 3.2-3 mesh and geometry deformation

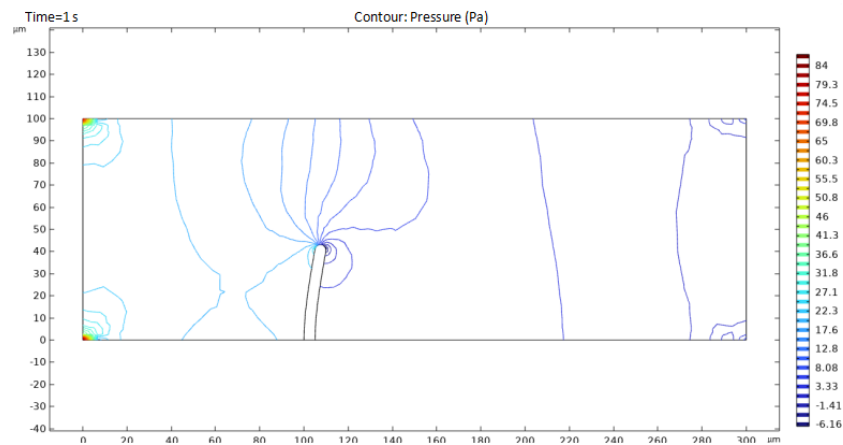


Figure 3.2-4 pressure results

From this replicated 2D simulation of the tutorial provided by Comsol Multiphysics, we can confirm that Comsol Multiphysics is capable of simulating fluid-structure interactions. This simulation also validates our capability to conduct this research and achieve our experimental objectives.

4. Theory

For this research, we conduct a one-side fluid-structure coupling to simulate the interaction between the aquaculture cage and the ocean current. When seawater flows through the cage, the deformation of the cage has little effect on the flow path and velocity distribution of seawater due to the high stiffness of the aquaculture cage, and thus can be ignored. In this scenario, the simplified form of the Navier - Stokes equation and the simplified continuity equation under the assumption of constant density are selected as the core control equations :

$$\rho \frac{Du}{Dt} = -\nabla p + \nabla \cdot \tau + \rho a \quad (1)$$

In this equation, $\frac{D}{Dt}$ is the material derivative, defined as $\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$, ρ is the (mass) density, $\mathbf{u}$ is the flow velocity, ∇ is the divergence, p is the pressure, t is time, τ is the deviatoric stress tensor, which has order 2, $\mathbf{a}$ represents body accelerations acting on the continuum, for example gravity, inertial accelerations, electrostatic accelerations, etc.

The continuity equation for fluid dynamics:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2) \quad 225$$

In this equation, ρ is fluid density, t is time, $\mathbf{u}$ is the flow velocity vector field. For seawater ρ is const; thus 226
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$$\nabla \cdot \mathbf{u} = 0 \quad (3) \quad 228$$

Navier - Stokes equation and the simplified continuity equation are widely used in CFD. Cao used these to simulate fluid-structure interactions of a rigid crustacean model with paddles performing metachronal movement, analyzing the effect of varied body shapes on propulsion efficiency [34]. 229
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Using these equations along with appropriate boundary conditions, we run simulations using Comsol and other software to obtain the results. 233
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5. Design and Analysis 235

5.1. Design of the System. 236

Semi-submersible rigid cages are equipped with a rigid framework that supports the photovoltaic system while enabling the system to control its height. They may provide the longest service life, as they are also widely used in other offshore industries [2]. This type of fish cage has a stable volume and keeps the fish within a certain range [35]. These features make it an ideal choice for the Integrated Photovoltaic and Fish Aquaculture System. 237
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However, this design has certain disadvantages. The semi-submersible fish cage requires a large capital investment. Due to its large, intricate steel structures, the cage is expensive to construct, demanding thorough engineering analysis, careful design, and strict quality control during construction to ensure safe offshore operation [2]. The high cost resulting from these factors remains a major drawback of the design. 242
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Examples of this cage design include Ocean Farm 1, Shenlan 1, Shenlan 2, and Viewpoint Seafarm etc. Ocean Farm 1, an integration of technologies and manufacturing techniques [2], was designed in Norway and manufactured in China. It has a diameter of 110 m and a volume of 250,000 m³, capable of carry over 1.5 million salmon. Shenlan 1, also a salmon farm, is located about 130 nautical miles off the coast of Shandong Province, China. With a diameter of 60 m and a height of 35 m, it can hold approximately 300,000 salmon. These numerous successful examples of semi-submersible rigid fish cages provide strong and convincing evidence that they would serve as an excellent base for the integrated photovoltaic and fish aquaculture system designed in this research. 247
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Our design of the photovoltaic and fish aquaculture system uses Shenlan 1 as a reference to design a semi-submersible rigid fish cage. To compare the different hydrodynamic responses caused by various cage shapes, we have chosen rectangular, octagonal, and hexadecagonal shapes. We will study and compare the different hydrodynamic outcomes resulting from these shape variations. 255
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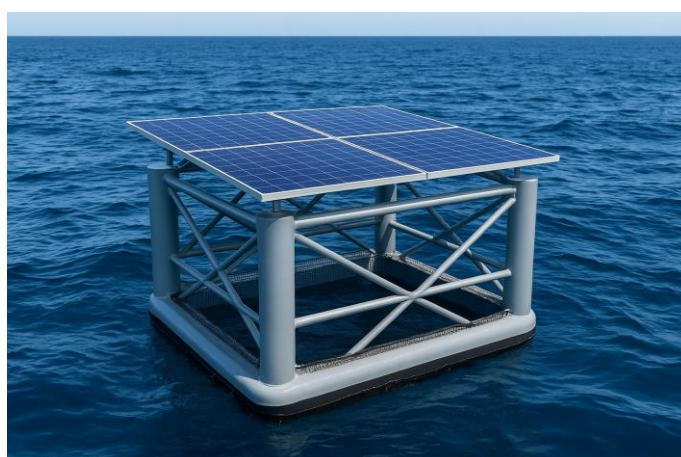


Figure 5.1-1 concept demonstration 1 (AI-generated)

For convenience in modeling and gridding the cages in multiphysics software, we simplify the model by excluding internal cavities and neglecting the structural thickness of the cage surface to simplify the problem. Therefore, in the simulations, we disregard the hollowness of the structural rods.

Concept demonstration are shown is Figure 5.1-1 and Figure 5.1-2 :

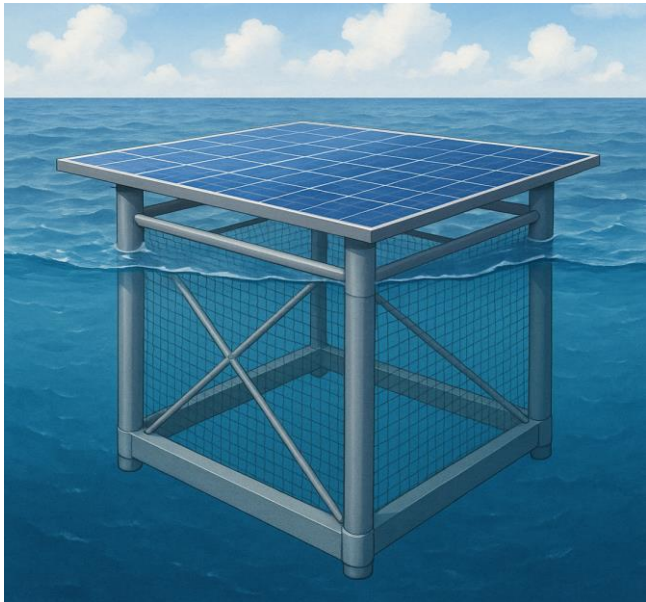


Figure 5.1-2 concept demonstration 2 (AI-generated)

5.1.1 Rectangular Shape

The rectangular cage is a rectangular prism-like structure with a square base, where the side length of the base is 53 m, the height is 38 m, and the volume is 406,742 m³. Other detailed data are listed below (Table 5.1.1-1), along with a demonstration figure (Figure 4.1.1-1).

Table 5.1.1 Parameters for rectangular cage

Member	Section(m)	Quantity	Length(m)	Thickness(m)
Bottom ring pontoon	B x H=4.0x5.0	4	53	0.028
Bottom corner pontoon	B x H=3.5x5.0	4	29.2	0.028
Corner column	D=3.5	4	38	0.028
Top ring brace	D=2.0	4	45	0.022
Top corner brace	D=1.5	4	30.2	0.015
Side brace	D=1.5	8	54.5	0.015

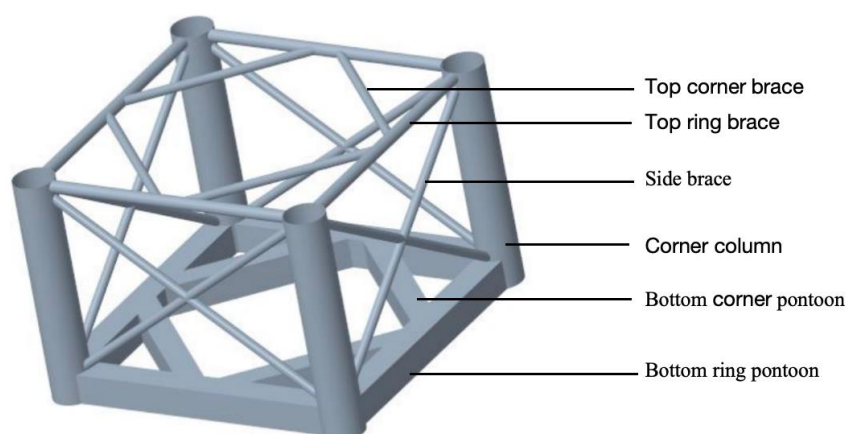


Figure 5.1.1 rectangular cage

5.1.2 Octagonal Shape

The octagonal cage design is a prism-like cage with an octagon base, where the side length of the base is 24 m and the height is 38 m, and has a volume of 105686 m³. Other detailed data are listed below (Table 5.1.2), along with a demonstration figure (Figure 5.1.2):

Table 5.1.2 Parameters for octagonal cage

Member	Section(m)	Quantity	Length(m)	Thickness(m)
Bottom ring pontoon	B x H=4.0x5.0	8	24	0.028
Bottom corner pontoon	B x H=3.5x5.0	4	27	0.028
Corner column	D=3.5	8	38	0.028
Top ring brace	D=2.0	8	22	0.022
Top corner brace	D=1.8	4	29	0.020
Side brace	D=1.5	8	36	0.015

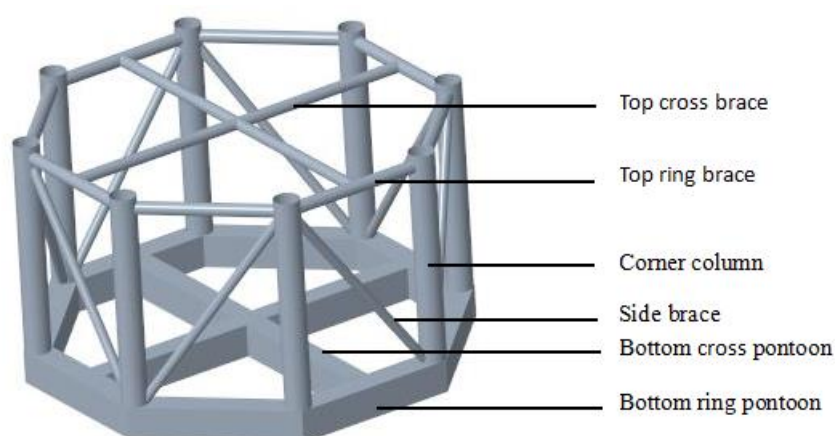


Figure 5.1.2 octagonal cage

5.1.3 Hexadecagonal Shape

The hexadecagonal cage design is a prism-like cage with a Hexadecagonal base, where the side length of the base is 12 m and the height is 38 m, and has a volume of 110039 m³. Other detailed data are listed below (Table 5.1.3), along with a demonstration figure (Figure 5.1.3):

Table 5.1.3 Parameters for hexadecagonal cage

Member	Section(m)	Quantity	Length(m)	Thickness(m)
Bottom ring pontoon	B x H=4.5x5.0	16	12	0.028
Bottom cross pontoon	B x H=3.5x5.0	4	28	0.028
Corner column	D=3.5	16	38	0.028
Top ring brace	D=2.0	16	10	0.022
Top cross brace	D=1.8	4	29	0.020

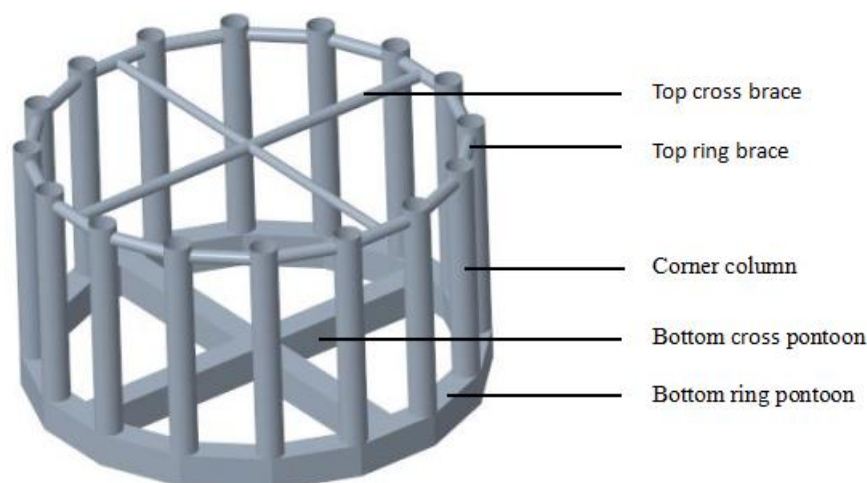


Figure 5.1.3 hexadecagonal cage

5.2 Setting

The requirements of offshore fishing consist of the following aspects. First, it should have a location at least 3km offshore with no surrounding shelter, and be fully exposed in the environment. Second, its location should have a depth of at least 50m or at least three times of the cage's height, with a minimum of 15m between the bottom of the cage and the seabed. Third, the speed of current at the location should be between 0.1 m/s and 1 m/s [2].

Among the factors, current speed is one of the most important for this study, as it is the main contributing factor to the water-cage interaction. An excessively high flow rate will have detrimental effects on the cage system and the fishes[2]. For salmon aquaculture, a current speed between 0.1 m/s to 0.6 m/s has been proven to be the most suitable [37].

The reference of the study, Shenlan 1, is located 130 nautical miles off the coast of Rizhao, in Shandong province, China, which we also consider an ideal reference condition. Therefore, the setup conditions and numerical values of the simulation of our project will consider the validity of the data in the given conditions of the location of Shenlan 1.

5.3 Experiment

5.3.1 Experiment with Rectangular Cage

First, we build a geometric structure of the fish cage into Creo and import it into Comsol Multiphysics. The model has a side length of 53m and a minimum rod diameter of 1.5m. As a rigid object, we ignore the hollow structure of the rods to simplify the simulations.

For the fluid domain, we construct a rectangular fluid domain with dimensions of 100m*200m*50m in Comsol to represent the seawater environment. This ensures the full development of the surrounding flow field and minimizes boundary effects.

For this simulation, we use Turbulent Flow, k- ϵ interface to model the fluid dynamics.

For the constants used, we set the density of the seawater to 1025 kg/m³ and the dynamic viscosity to 0.001 Pa·s. For the boundary conditions, based on the current data from Yellow Sea near Dalian, we have set the uniform velocity inlet with a flow speed of 0.5 m/s [36]. The pressure outlet is set to be 0 Pa. And we treat all surfaces of the cage to be no-slip walls, indicating that the surface velocity of the fluid on the contact surface would be 0 m/s. We set the rectangular fluid domain to have slip walls, simulating the domain in ocean environments.

Next, we apply six layers of boundary layers to generate mesh to resolve near-wall effects accurately.

In the simulation, we will conduct a steady-state analysis, as mentioned before, and default Comsol setting were used.

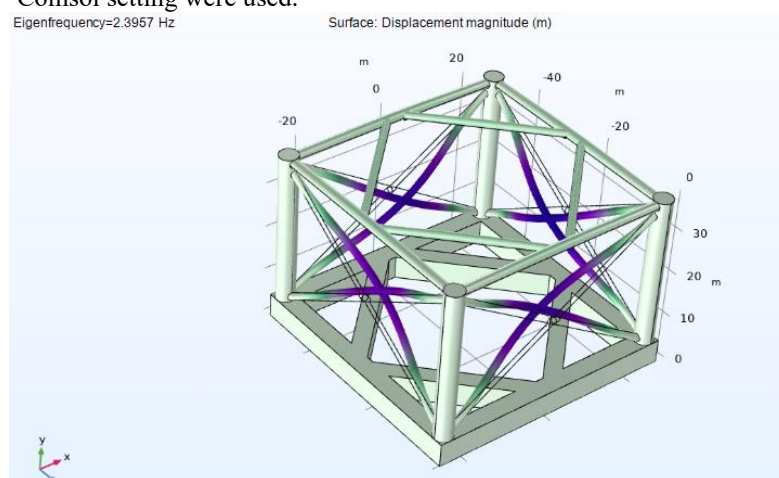


Figure 5.4.1-1 Eigen frequency and surface displacement magnitude

When the fish cage is in a hydrostatic equilibrium state, ballast is primarily distributed across the bottom ring beams, corner struts, side columns, and corner posts. In this state, the system behaves like a monopile floating body and is unconditionally stable. The material density: 7.85×10^3 kg/m³. Structural volume: 8.6×10^3 m³. Natural frequency: 2.39 Hz, significantly higher than typical ocean wave frequencies, allowing the cage to be treated as a rigid body in the analysis (as shown in Figure 5.4.1-1).

For the conditions in this simulation, according to relevant data from the North Yellow Sea, China, we assume the water depth is 50–100 m and the surface current velocity is 0.3–0.5 m/s [36]. We also assume the average wave height in summer is 1.0–1.5 m. Based on these conditions, a k- ϵ turbulence model was applied in the simulation, yielding the following representative results.

The wall resolution in viscous units is shown in Figure 5.4.1-2. And surface pressure distribution is shown in Figure 5.4.1-3.

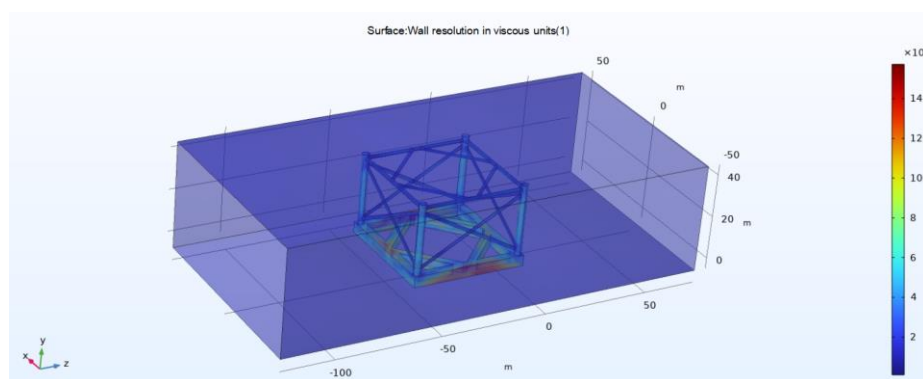


Figure 5.4.1-2 viscous units

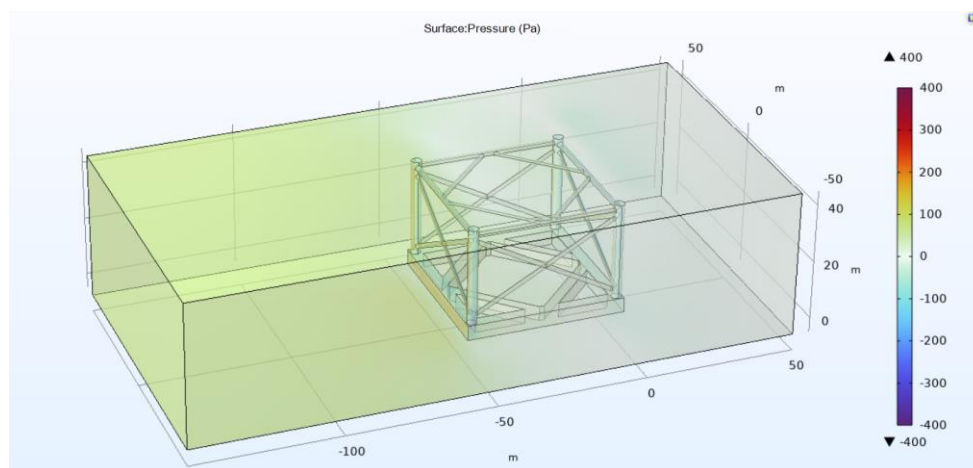


Figure 5.4.1-3 surface pressure

Internal Flow Velocity: Vector fields or streamline plots were extracted from sectional or volumetric data to depict internal flow velocities. Comparisons between internal and external flow velocities were also conducted (see Figure 5.4.1-4, Figure 5.4.1-5).

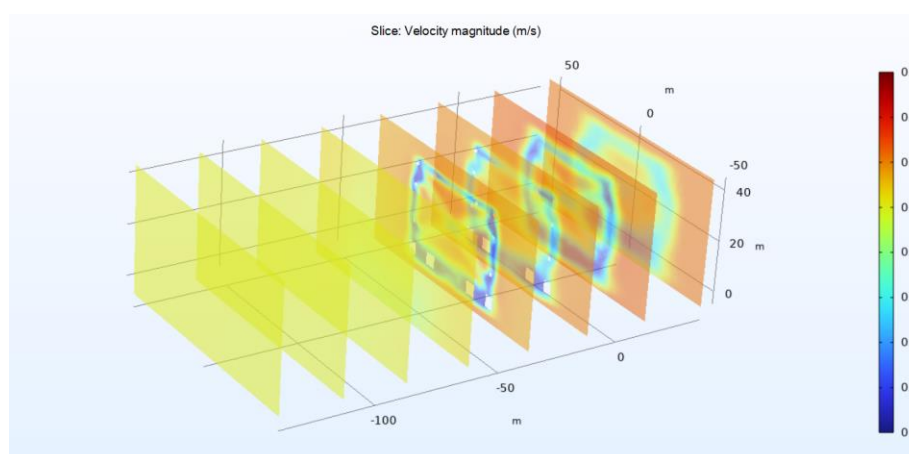


Figure 5.4.1-4 velocity magnitude in XZ

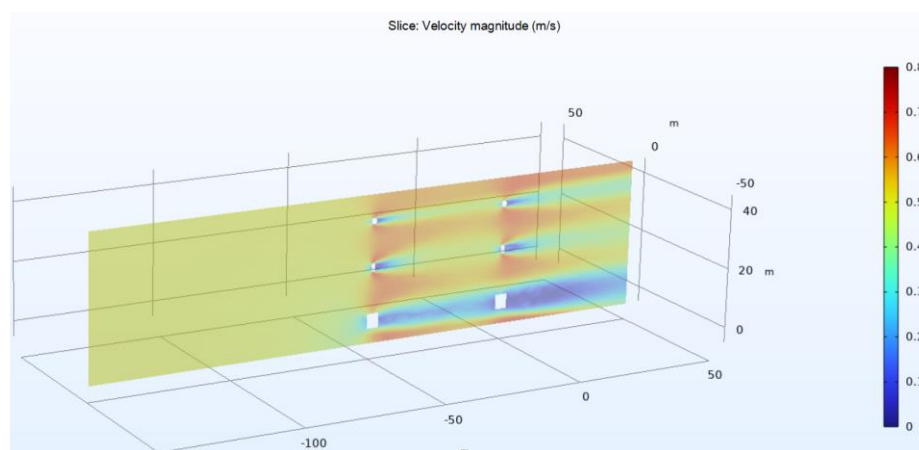


Figure 5.4.1-5 velocity magnitude in YZ

5.4.2 Experiment with Octagon Cage

Using the same set of conditions and numerical figures, we have performed the experiment with the octagonal cage as well (see Figure 5.4.2-1, Figure 5.4.2-2, Figure 5.4.2-3).

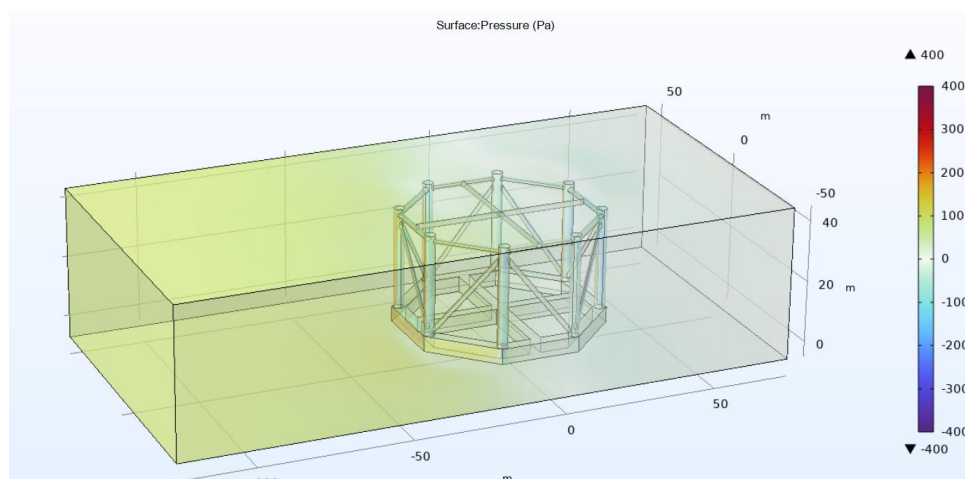


Figure 5.4.2-1 surface pressure

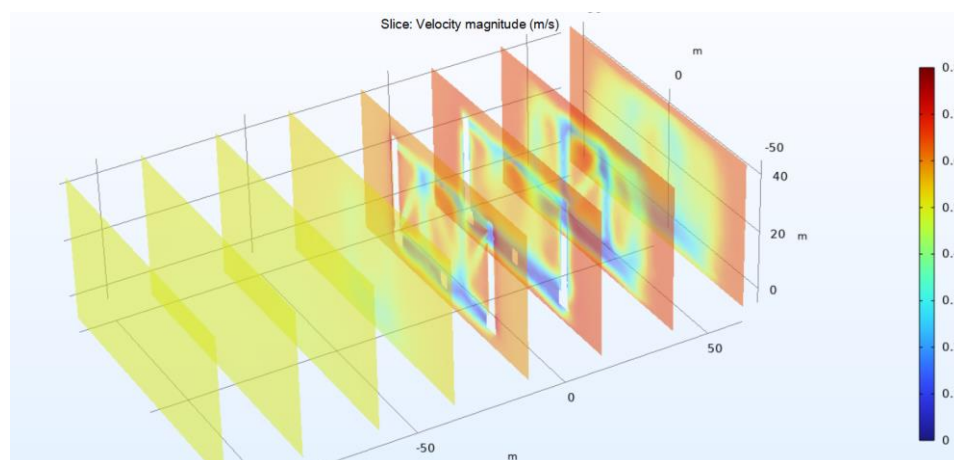


Figure 5.4.2-2 velocity magnitude in XZ

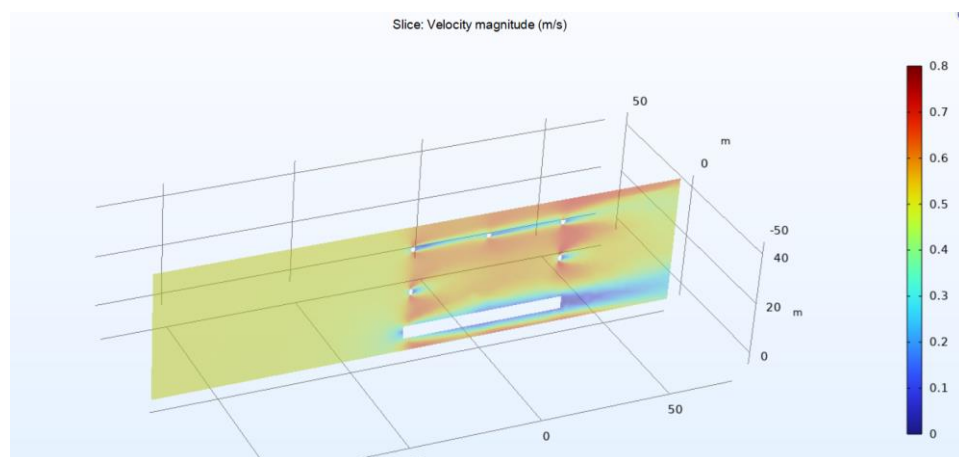


Figure 5.4.2-3 velocity magnitude in YZ

5.4.2 Experiment with Hexadecagonal Cage

Using the same set of conditions and numerical figures, we have performed the experiment with the hexadecagonal cage as well (Figure 5.4.3-1, Figure 5.4.3-2, Figure 5.4.3-3).

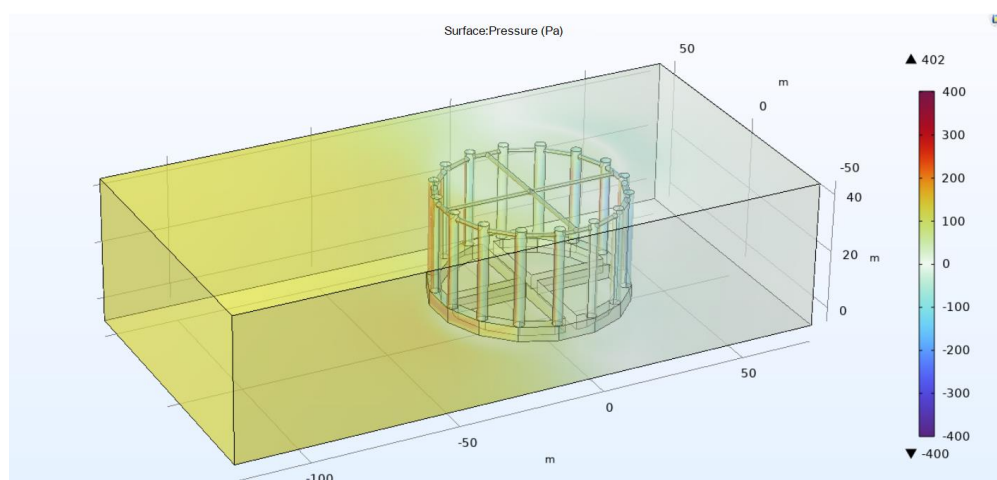


Figure 5.4.3-1 surface pressure

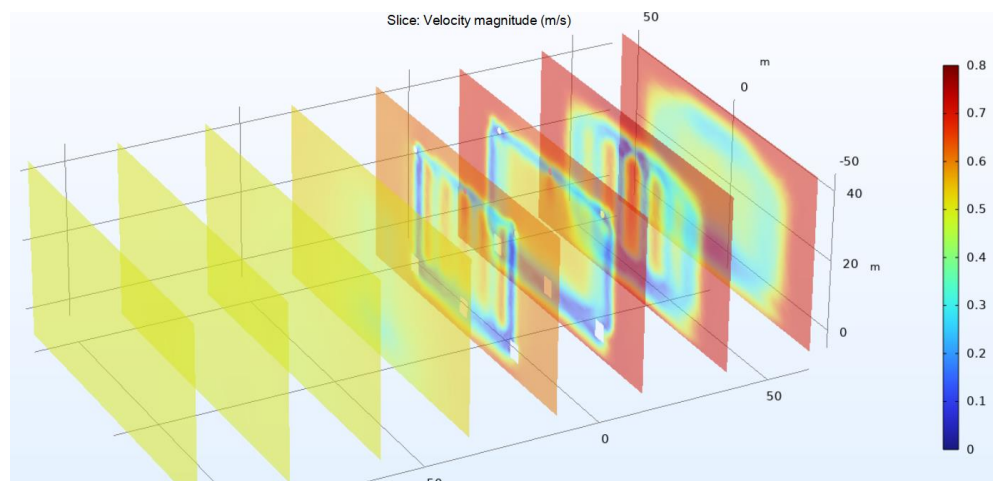


Figure 5.4.3-2 velocity magnitude in XZ

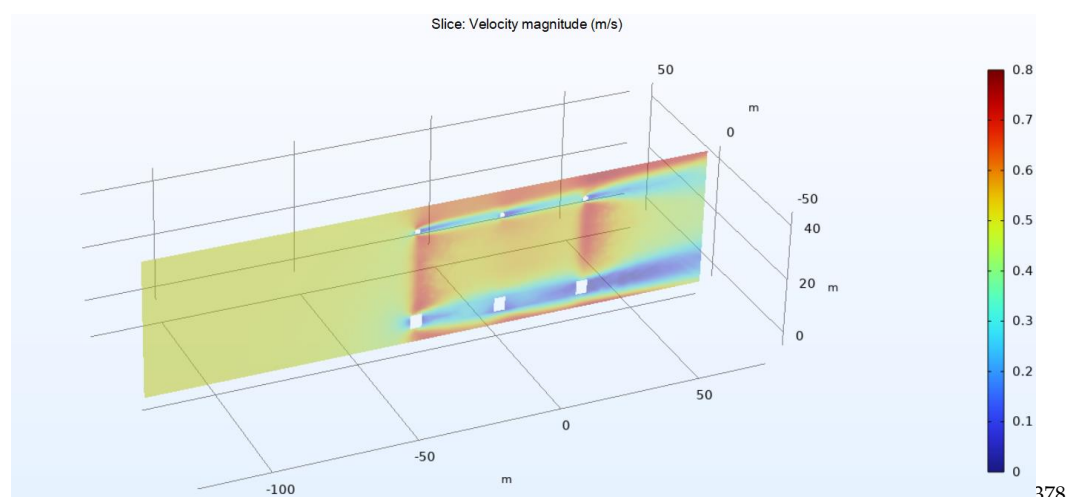


Figure 5.4.3-3 velocity magnitude in YZ

6. Results and discussions

6.1 Surface Pressure Distribution

In Figures 5.4.1-3, 5.4.2-1, and 5.4.2-1, clear evidence indicates that the pressure distributed on the upstream surface increases along the side number of the cage design. The pressure directly exerted on the cage surface, ranging from 100 Pa, 120 Pa to 140 Pa, also shows an increase along the increase of the number of sides. Overall, it can then be confirmed that the increase in the side numbers have a direct effect on the pressure distribution, with the hexadecagonal cage having the highest pressure and structural demands, and the rectangular cage the lowest.

This can be studied by analyzing how seawater flow is resisted by the cage. When the seawater flows through the cage, the pressure difference between the front and rear areas will form the pressure difference resistance. Meanwhile the frictional resistance caused by the viscous friction between the water flow and the cage surface. When the number of cage sides increases, the blocking area or structural disturbance of water contact increases, so the differential pressure resistance and friction resistance will increase; As a result, the pressure of seawater on the cage is increasing.

We can also notice that the diagonal support rod design significantly reduces flow velocity and improves fish welfare. The diagonal support rods form a great resistance to the flowing fluid, which can significantly reduce the flow speed and share the pressure brought by the fluid.

6.2 Internal Flow Velocity

Figure 5.4.1-4, Figure 5.4.2-2, Figure 5.4.3-2, Figure 5.4.1-5, Figure 5.4.2-3, Figure 5.4.3-3 are flow velocity graphs where flow velocity is extracted as vector fields or streamline plots from sectional or volumetric data. Therefore comparisons of internal and external flow velocities can be made. It is evident that there is a significant decrease in internal velocity along the increase of the number of sides, with hexadecagonal cages having the slowest internal flow velocity, while the rectangular cage shows the highest internal flow velocity. The hexadecagonal cage has a fluid velocity of about 0.53 m/s, while the rectangular cage has a fluid velocity of about 0.6 m/s. It can also be noticed that the diagonal support rod has a great influence on the flow speed, decreasing the flow velocity from 0.6 m/s to nearly 0.5 m/s for the rectangular cage, an approximately 15% decrease in the flow velocity. Additionally, the hexadecagonal cage has the lowest flow velocity in the wake region, while the rectangular cage shows the highest. The difference in the flow velocity outside the cages is due to the fluid domain and is a computational artifact which should be excluded from analysis.

From the graphs, we can notice that there is a clear correlation between internal fluid velocity and the number of sides across multiple designs of the integrated photovoltaic and fish aquaculture system under identical initial conditions. A clear decrease of the internal fluid velocity is manifested as the number of sides increases, as shown in the figures. The correlation is crucial towards future

cage designs, for it provides a systematic approach to adjust the inner flow speed of the fluid in the cage to better suit the need of fish welfare. This provides a more stable and safer environment for fish movement and muscle development.

7. Conclusions

Under the same initial conditions and simulated ocean environments, the hexadecagonal cage exhibits the lowest internal flow speed, making it the optimal option in terms of fish welfare and aquaculture performance. For salmon farming, the most ideal flow speed range is 0.1–0.6 m/s [37]. It could be said the hexadecagonal cage design is well-suited to this requirement from this perspective. However, it incurs higher material and manufacturing costs, and it is more challenging to scale into a large solar farm. Additionally, its multi-sided design complicates assembly. The hexadecagonal cage also exhibits the highest surface pressure, imposing greater structural and material demands and reducing the facility's long-term durability. Therefore, with advancements in material engineering, the hexadecagonal cage design could emerge as one of the best among the three.

While the rectangular cage is a classic design, it has the highest internal fluid speed. Although it is less ideal than the hexadecagonal design, it experiences significantly lower surface pressure. Moreover, it is easier to assemble into large-scale solar farms consisting of numerous integrated photovoltaic and fish aquaculture units. The diagonal support rod design provides sufficient structural support and notably reduces internal flow velocity in the wake of these rods, a feature that enhances its competitiveness.

The octagonal cage design shows no distinct advantages or defining features. It falls between the other two designs in nearly all metrics measured in this experiment. Given its relatively high manufacturing costs and assembly difficulties, it offers no clear advantages over the other two designs.

Overall, this study concludes that increasing the number of sides in the cage design better supports fish welfare because aquaculture typically requires a low-flow environment. Moreover, more sides lead to the need for higher structural strength due to increased pressure distribution. Another novel finding is that the diagonal support rod design significantly reduces flow velocity and improves fish welfare. Additionally, increasing the number of sides lowers the flow velocity behind the cages, enhancing their potential for assembly.

8. Future Works

This work is not a comprehensive study of the design of an integrated aquaculture system. The research solely focuses on the effects of the number of sides. The performance of other possible cage designs and the possibility and efficiency of assembly, as well as the efficiency in space management, warrants further investigation. In the future, structural analysis for manufacturing and the relationship between the power supply from the photovoltaic facilities and the power demand of the system will also need consideration.

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